

Rishikesh Bastakoti

Newark, New Jersey

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Education

Caldwell University — GPA: 3.90/4.0

Bachelor of Science in Computer Science

Aug 2024 - May 2028

Caldwell, NJ

Relevant Coursework: Computer Programming I and II (Python), Database Management Systems, Windows Programming, Discrete Structures, Operating Systems, Data Structures, Calculus I, Design and Analysis of Algorithms

Awards and Honors: Presidential Scholarships, St. Thomas Scholarships, Dean's List (2024, 2025)

Technical Skills

Languages: Python, JavaScript/TypeScript, Rust (via PyO3), SQL, C, C#(Basic)

AI & Machine Learning: LangChain, Ollama (Llama 3.1/3.3), Semantic Embeddings (nomic-embed-text), NLP (Cosine Similarity/Regex), Isolation Forest (Anomaly Detection), NumPy, Pandas

Backend & Cloud: FastAPI, Streamlit, FastHTML, Node.js, Vercel Serverless, RESTful APIs, JSON

Databases: Neo4j (Graph DB), PostgreSQL, Supabase, SQLite, MySQL

Tools & Architecture: Git, GitHub, Docker, Tailwind CSS, SQLAlchemy, JWT Authentication

Personal Projects and Research

Behavior-Aware Personal Budget Tracker (Undergraduate Research)

Developed a full-stack financial app (Python, FastAPI, Streamlit) for undergraduate research on FinTech's "Explainability Gap." It utilizes a deterministic Rule Engine to analyze simulated Pandas profiles, automatically flagging triggers like "Subscription Creep." An Isolation Forest machine learning algorithm, paired with regression threshold logic, isolates the top 1.5% of erratic transactions to translate raw statistical anomalies into actionable coaching insights.

GitHub: github.com/reseekesh821/BudgetTracker

AI Compliance Firewall (LLM Guardrails Middleware)

Engineered a high-throughput LLM security proxy utilizing FastAPI and FastHTML to enforce HIPAA and FINRA compliance rules on generative AI interactions. The system features a multi-tier detection pipeline that combines native regex rules with an Ollama embedding layer (nomic-embed-text) to catch semantic paraphrasing via cosine similarity. Additionally, it integrates a Neo4j knowledge graph to dynamically inject or redact required legal disclaimers into Llama 3.1 8B outputs, logging all system actions to a local SQLite audit database. Processing speeds were further accelerated by compiling a custom pattern-matching rule engine natively in Rust using Maturin and PyO3.

GitHub: <https://github.com/reseekesh821/AIComplianceFirewall>

AI-Powered Portfolio Platform

Built a production-ready portfolio using Vercel Serverless Functions to host edge-optimized AI capabilities. It features a Groq-backed (Llama 3.3 70B) chatbot capable of executing DOM manipulations—like controlling media—via native browser speech recognition. A scalable Supabase backend cluster securely manages chat logs, site analytics, and multi-lingual user events in real-time.

GitHub: <https://github.com/reseekesh821/my-portfolio>

Live: <https://rishikeshbastakoti.vercel.app>

Participations

Cougar Health – Mental Wellness WebApp (Team Project for Cougar Hacks 2025 Hackathon)

Collaborated on a full-stack web app using Next.js, TypeScript, Prisma, and PostgreSQL to support students struggling with depression, suicide, and drug abuse. Key features include AI chat support (Groq + LLaMA), quiz-based rewards, anonymous reporting, and survivor stories.

GitHub: [GitHub Repo](#)